AirTransformer freedom to operate status

Savery Tech | 3 November 2026 | EIC Accelerator C05

Owned technology and product scope

Savery owns the system patent originating from its 2019 filing. The identified family includes European application EP19306516.6, publication EP3825627A1, and PCT/EP2020/082838, publication WO2021099546A1. These publications concern cascaded adsorption/desorption cooling and heating. The current AirTransformer product follows the CAU-series material route and combines sorption air treatment, thermal management, cyclic operation and indirect evaporative cooling. Savery controls the system architecture and engineering know-how. Its protection strategy also addresses process instructions, calibration, firmware, models and manufacturing-transfer information. [1]

The assessment scope follows the manufactured and deployed product: selected material composition and preparation, shaping and coating, substrate and exchanger geometry, air/heat/water circuits, operating sequence, control software, installation and service. Activities and countries are considered separately. The immediate scope is European development, manufacture and deployment. North American and Asian production or sales require review of the specific countries and applicable family members before entry.

Analysis already available

Plasseraud's 29 January 2025 HMX patentability analysis considered the two-MOF exchanger and related material use against seven documents. It distinguished material and substrate claims from indirect evaporative geometries and liquid-desiccant combinations, and identified the need to substantiate the specific arrangement and engineering benefits. This work supports protection and claim drafting. It does not constitute a completed country-specific FTO opinion for every current product version. [2]

The updated product screen retains those distinctions and adds the following published families. The comparison concerns actual technical features, rather than treating every MOF patent as a claim on all MOF use.

Screening lead	Relevant feature and product comparison
BASF, US11609000B2	Claims 1, 5 and 12 concern the specified sorption/heat-exchanger channel structure and associated process/apparatus. Compare actual contactor channels, coating distribution and air/thermal routing. Changing the material name alone does not settle the comparison. [3]
NuMat, EP4598672A1 / WO2024107531A1	Claim 1 addresses hybrid granules containing two adsorbent materials and a binder, with the stated spatial relationship. Compare the selected coating or shaped-material recipe and morphology; the granule structure is a specific screening condition. [4]
Framergy and Absorption Solutions, EP3890878A2 / US20220023829A1	The published subject matter concerns amorphous MOFs and preparation on porous substrates. Compare crystalline CAU identity, processing and substrate formation. Review national prosecution and applicable claims separately; an international or regional database label is not a worldwide family conclusion. [5]
Montana/AirJoule, US20250043971A1	Claim 1 includes thermally coupled contactors in chambers capable of sealed and unsealed states. Compare the adopted regeneration pressure, chamber sealing, contactor coupling and water-recovery arrangement. A shared low-temperature heat range is not itself the decisive claim feature. [6]

The earlier HMX analysis also identified US11786883B2, an adsorption-cooling/substrate family of Lawrence Livermore National Security, and indirect-cooling documents US9140471B2 and WO2020252059A1 associated with Alliance/NREL. These remain separate leads; they are not attributed to the companies above. [2]

Configuration release and market entry

The next decision is a professional claim-by-claim review of the released bill of materials, drawings and manufacturing method against the applicable national rights. It will record each material feature or process step, the relevant claim element, territorial status, the proposed activity and the disposition. An owned system patent and a third-party FTO conclusion serve different purposes. This note records the present assessment and release process; it does not certify global clearance.

Frédéric Cui owns the product configuration decision and coordinates the assessment with specialist IP support. The productisation programme budgets IP and compliance work alongside design and manufacturing release. Any identified issue will be resolved through a documented technical change, appropriate permission or an adjusted

territory plan before the affected commercial activity. Changes to materials, shaping, heat-exchanger geometry, regeneration, suppliers or intended uses trigger a proportionate update.

Employee, contractor and collaboration arrangements will preserve Savery's rights to the results it needs for manufacture, modification, maintenance and service. Controlled disclosure, confidentiality and publication review protect process knowledge and patentable improvements. Supplier transfers will define the permitted use of drawings and instructions, the ownership of improvements and the access needed for continuing production and replacement supply. Data and software records will identify background dependencies, licences and operational-data permissions. These measures connect the IP strategy to the product that will be manufactured and supported.

References

[1] Company management ownership and product confirmation, 3 November 2026; EP3825627A1 and WO2021099546A1 publications. [2] Plasseraud IP, Analyse de brevetabilité HMX, B2500097EP, 29 January 2025, pp. 1–3. [3] https://patents.google.com/patent/US11609000B2/en [4] https://patents.google.com/patent/EP4598672A1/en [5] https://patents.google.com/patent/EP3890878A2/fr [6] https://patents.google.com/patent/US20250043971A1/en